Mapping Chester

Vectors by Hand

This week you will learn to split a vector into horizontal and vertical parts, add vectors, and find where you end up. You will finish by drawing a route across Chester and finding your displacement from Subaru Park at every stop.

Plan your week

Write the day you plan to do each part. Aim for about **45 minutes a day**. The whole packet takes about **4 hours**, so don't save it all for Friday.

Part	What you'll do	About	Day I'll do it	Done
1	Warm-up: directions and signs (no calculator)	15 min		
2	Pythagorean theorem	25 min		
3	SOHCAHTOA: breaking a vector apart	35 min		
4	Adding vectors	30 min		
5	Finding direction	20 min		
6A–6B	The Chester history route (map + table)	60 min		
6C	Your own route to school	40 min		
	Reflection + photos + submit	15 min		

Tips for a busy week

- Do the parts in order. Each part uses the one before it.
- Problems marked ★ **Challenge** are optional. Skip them if you are short on time.
- Stuck for more than 10 minutes? Circle the problem, write one sentence about what you tried, and move on. Message your teacher.

What you need

Item	Used for
Ruler with centimeters	Drawing vectors to scale and measuring them
Protractor	Drawing and measuring angles
Drawing compass	Checking lengths with arcs and circles
Scientific calculator in DEGREE mode	sin, cos, tan and $\tan^{-1}$. Check: sin(30) should equal 0.5
Pencil + 2 colored pencils	One color for components, one for resultants
A map of Chester (phone or printed)	Part 6C only

Printing check. Print this packet at 100% ("Actual size," not "Fit to page") so grid squares are 1 cm.



This bar should measure exactly 10 cm with your ruler.

The rules. Work by hand, in pencil. For every calculation show three things: the formula, the numbers plugged in, and the answer with units. When a problem says *measure*, use your ruler or protractor and write what you measured next to what you calculated. They should be close, not identical.

Part 1 · Warm-up: Which way is that?

About 15 min · no calculator

Short answers are fine. Sketch inside the boxes.

1.1 In your own words, what is the difference between **distance** and **displacement**? Give one example from your own day.

1.2 Draw a compass rose. Label N, E, S, W, NE, SE, SW and NW. How many degrees are between N and E? Between N and NE?

N to E = _____° N to NE = _____°

1.3 **"30° N of E"** means: face east, then turn 30° toward north. With your protractor, draw an arrow pointing 30° N of E. Then draw one pointing 30° E of N. Are they the same direction?

Same direction? YES / NO

1.4 A Union midfielder at Subaru Park passes the ball diagonally upfield. Sketch the field from above and the ball's path. Add two dashed arrows: one for the **across-the-field** part and one for the **toward-the-goal** part. Which part brings the ball closer to the goal?

1.5 In this activity, **east and north are positive (+)**, and west and south are negative (−). Write the sign for each: 4 km west ____ 2 km north ____ 7 m south ____

1.6 **Predict** (no math): you walk 3 blocks east, then 4 blocks north. Is your straight-line distance from the start **more than 7**, **exactly 7**, or **less than 7** blocks? Why?

Part 2 · Two perpendicular vectors

About 25 min

Big idea from Part 1: any diagonal vector can be replaced by a horizontal piece plus a vertical piece. Those pieces are called **components**.

When two vectors are at 90° to each other, they make the two legs of a right triangle. The **resultant** R, the single arrow from start to finish, is the hypotenuse.

$$R = \sqrt{(x^2 + y^2)}$$

Worked example. A Union winger runs 30 m north along the sideline, then cuts 40 m east.
Formula: $R = \sqrt{(x^2 + y^2)}$ → Plug in: $R = \sqrt{(40^2 + 30^2)} = \sqrt{(1600 + 900)} = \sqrt{2500}$ → **R = 50 m**
Drawing check: at 1 cm = 5 m, the legs are 8 cm and 6 cm, so R should measure 10 cm.

For 2.1 to 2.3: **(a)** draw it tip-to-tail on the **Part 2 graph paper** (next page), **(b)** calculate R here, **(c)** measure R with your ruler and convert it with the scale.

2.1 You walk 3 blocks east, then 4 blocks north. Scale: 1 cm = 1 block. Was your prediction in 1.6 right?

Work

Calculated R = _____ Measured R = _____

2.2 A bus leaves the Chester Transportation Center and drives 1.2 km east, then 0.5 km south. Scale: 1 cm = 0.1 km.

Work

Calculated R = _____ Measured R = _____

2.3 A delivery bike rides 2.0 km east, then 3.0 km north. Scale: 1 cm = 0.5 km. Round R to 2 decimal places.

Work

Calculated R = _____ Measured R = _____

2.4 Compass check. Open your drawing compass to the length of your arrow R from 2.3. Put the point on the start and swing an arc. Does it pass through the finish? If not, find and fix your error.

Did it pass through? YES / NO If no, what did you fix? _____

2.5 Working backwards. A drone flies 13 m in a straight line. Its east component is 5 m. How far north did it go? Sketch it. Hint: $y = \sqrt{(R^2 - x^2)}$.

Work

North = _____

Part 2 drawings (problems 2.1, 2.2, 2.3)

Name: _____ Write each problem's scale next to its drawing

Each small square = 0.5 cm. Each bold square = 5 cm. Print at 100% ("Actual size").

Part 3 · Breaking a vector apart: SOHCAHTOA

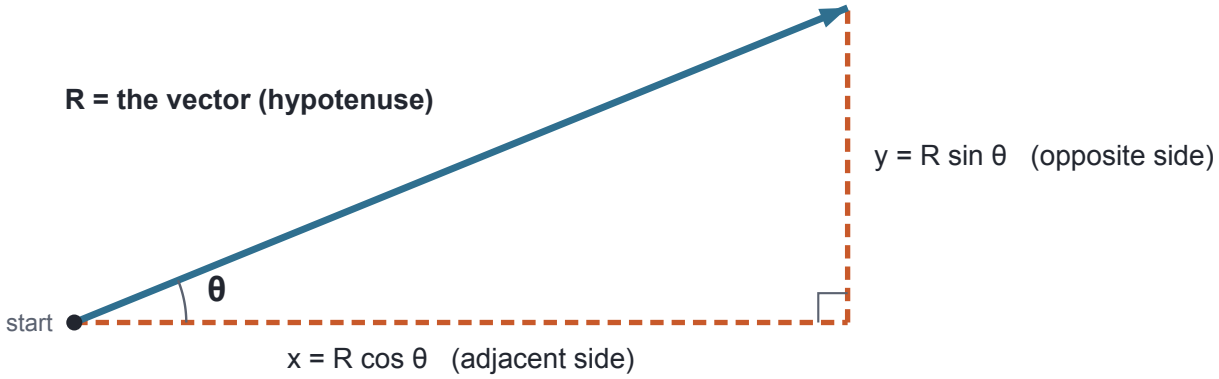
About 35 min

If you know a vector's size and angle, trig gives you its components. Draw the vector as the hypotenuse, then draw a horizontal and a vertical line to make a right triangle.

SOH	CAH	TOA
Sine = **O**pposite ÷ **H**ypotenuse	**C**osine = **A**djacent ÷ **H**ypotenuse	**T**angent = **O**pposite ÷ **A**djacent

When the angle θ is measured from the east–west (horizontal) axis:

$$x = R\cos\theta \qquad y = R\sin\theta$$

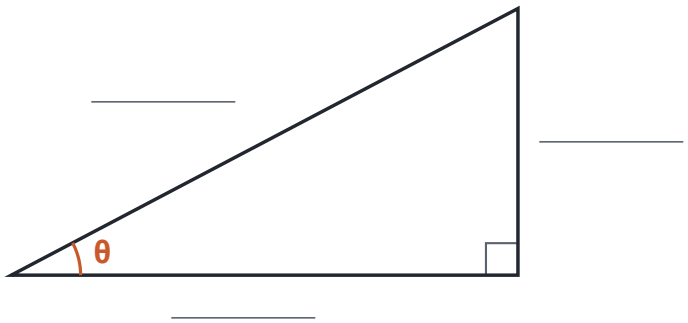


Watch the angle. If the angle is measured from the north–south axis (like "25° W of N"), sine and cosine switch places. Always ask: which side is *next to* my angle? That one uses cosine.

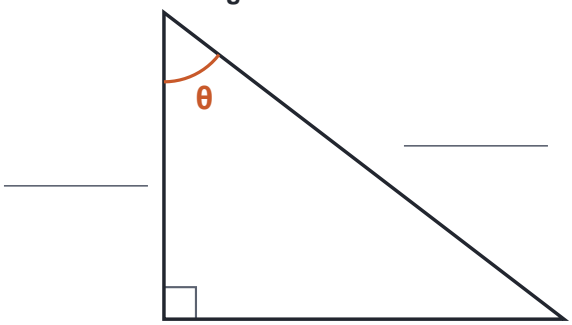
Worked example. A goalkeeper punts the ball at 20 m/s, 30° above the ground.
Horizontal: x = 20 cos 30° = 20 × 0.866 = **17.3 m/s** Vertical: y = 20 sin 30° = 20 × 0.500 = **10.0 m/s**
Check: √(17.3² + 10.0²) = 20.0 m/s ✓

3.1 On each triangle, write **opposite**, **adjacent** or **hypotenuse** on the blank next to each side. Use the angle θ that is marked.

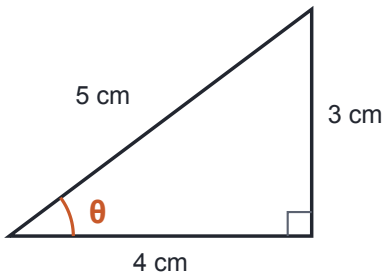
Triangle A



Triangle B



3.2 Use this 3-4-5 triangle. Write sin θ, cos θ and tan θ as fractions, then as decimals.



Printed to scale. Measure it with your ruler to check!

sin θ = _____ = _____ cos θ = _____ = _____ tan θ = _____ = _____

3.3 A student walks 10 m at 30° N of E. **(a)** Draw it on the Part 3 graph paper (1 cm = 1 m). **(b)** Measure the x and y components with your ruler. **(c)** Calculate them with cos and sin.

	Measured	Calculated (show work below)
x (east)		
y (north)		

Work

3.4 A free kick leaves the foot at 18 m/s, 35° above the ground. Sketch the triangle. Find the horizontal and vertical components.

Sketch + work

horizontal = _____ vertical = _____

3.5 ★ **Challenge (optional).** A hiker walks 4.0 km at 25° W of N. Sketch it. Which component uses sine, and which uses cosine? Find both with the correct signs.

Sketch + work

x = _____ y = _____

Part 3 drawings (problem 3.3)

Name: _____ Scale: 1 cm = 1 m

Each small square = 0.5 cm. Each bold square = 5 cm. Print at 100% ("Actual size").

Part 4 · Adding vectors

About 30 min

Use both methods on each problem. The drawing checks the math, and the math checks the drawing.

Method 1: Tip-to-tail (draw). Draw the first vector. Start the next vector at the *tip* of the last one. The resultant goes from the very first tail to the very last tip.

Method 2: Components (calculate). (1) Break every vector into x and y. (2) Add all the x's to get R_x and all the y's to get R_y. Keep the signs! (3) $R = \sqrt{(R_x^2 + R_y^2)}$.

> **Worked example.** A = 3.0 km east. B = 2.0 km at 60° N of E.
> A: x = 3.0, y = 0 B: x = 2.0 cos 60° = 1.00, y = 2.0 sin 60° = 1.73
> R_x = 3.0 + 1.00 = 4.00 km R_y = 0 + 1.73 = 1.73 km $R = \sqrt{(4.00^2 + 1.73^2)}$ = **4.36 km**

Draw every problem on the **Part 4 graph paper**. Keep your answers: you'll find their directions in Part 5.

4.1 **Same line.** No trig needed. (a) 5 m E + 3 m E = _____ (b) 5 m E + 3 m W = _____ (c) 2 km N + 6 km S = _____

4.2 **Does order matter?** Draw 4 km E then 3 km N, tip-to-tail. From the same start, in a different color, draw 3 km N then 4 km E. What do you notice about the two resultants?

4.3 A = 5.0 m at 30° N of E. B = 4.0 m north. Fill in the table, then find R. Draw it (1 cm = 1 m) and measure R.

Vector	x component (with sign)	y component (with sign)
A		
B		
Resultant	R_x =	R_y =

Work

Calculated R = _____ Measured R = _____

4.4 A walk around Chester: 2 km north, then 3 km east, then 1 km south. Find R by components, then by drawing (1 cm = 1 km).

Vector	x component (with sign)	y component (with sign)
2 km N		
3 km E		
1 km S		
Resultant	R_x =	R_y =

Work

Calculated R = _____ Measured R = _____

4.5 ★ **Challenge (optional).** A = 6.0 km at 40° N of E. B = 3.0 km at 20° S of E. Careful: B's y component is negative. Draw at 1 cm = 0.5 km.

Vector	x component (with sign)	y component (with sign)
A		
B		
Resultant	R_x =	R_y =

Work

Calculated R = _____ Measured R = _____

Part 4 drawings (problems 4.2 to 4.5)

Name: _____ Write each problem's scale next to its drawing

Each small square = 0.5 cm. Each bold square = 5 cm. Print at 100% ("Actual size").

Part 5 · Which way? Finding direction

About 20 min

A resultant isn't finished until it has a direction. Use inverse tangent ($\tan^{-1}$ on your calculator) with the **sizes** of the components. Ignore the signs for this step:

$$\theta = \tan^{-1}(|R_y| \div |R_x|)$$

This gives the angle from the **east–west axis**. Then use the signs to name the direction:

R_x is	R_y is	Write the direction as
+	+	θ N of E
−	+	θ N of W
−	−	θ S of W
+	−	θ S of E

Worked example. From Part 4: R_x = 4.00 km, R_y = 1.73 km. $\theta = \tan^{-1}(1.73 \div 4.00) = 23.4°$. Both are positive, so: **4.36 km at 23.4° N of E**.

For each row, calculate θ, then **measure it with your protractor** on your earlier drawing. Show your $\tan^{-1}$ work in the box below.

#	From	R_x	R_y	θ calculated	Direction (__ of __)	θ measured
5.1	Problem 2.1					
5.2	Problem 2.2					
5.3	Problem 4.3					
5.4	Problem 4.4					
5.5 ★	Problem 4.5 (optional)					

Work

5.6 A vector has R_x = −3 m and R_y = −3 m. **Without a calculator**, give its direction. How did you know?

Part 6 · Mapping Chester

About 100 min · split over 2 days

You will draw a route through Chester's history one leg at a time and track your displacement from Subaru Park after every stop. Then you'll map a route of your own.

6A · Get your map ready (5 min)

Find the **Chester route map sheet** in this packet. It already has the start (Subaru Park), a north arrow and the scale: **1 cm = 0.25 km**, so **1 km = 4 cm**. Leg 1 is drawn for you in gray as an example.

6B · The Chester history route (about 60 min)

A class trip starts at Subaru Park and visits four stops. Each leg is a straight-line ("as the crow flies") displacement. Stop positions are simplified for this activity, so they are close to, but not exactly, the real locations.

Leg	From → To	Displacement
1	Subaru Park → the **Penn Landing site** on the waterfront, where William Penn first came ashore in Pennsylvania in 1682	1.80 km at 34° N of E
2	Penn Landing site → the **1724 Chester Courthouse**, one of the oldest public buildings in the country	0.85 km at 69° N of E
3	1724 Courthouse → **Widener University**	1.41 km at 82° N of E
4	Widener University → the former **Crozer Theological Seminary** in Upland, where Dr. Martin Luther King Jr. studied from 1948 to 1951	1.97 km at 15° S of W

For each leg, in order:

1. **Convert** the length to cm: km ÷ 0.25 = cm.
2. **Draw** the leg with your protractor and ruler, starting at the tip of the last leg. Label the stop.
3. **Calculate** the x and y components with SOHCAHTOA. Watch the signs on leg 4: it heads south and west.
4. **Add** this leg's x to your running R_x, and its y to your running R_y.
5. **Calculate** your displacement from Subaru Park (Pythagorean theorem) and its direction ($\tan^{-1}$).
6. **Draw** a dashed arrow, in a second color, from Subaru Park to the stop you just reached.

> Leg 1 is completely worked out for you on the next pages: in the data table, in the work box, and on the map. Use it as your model for legs 2 to 4.

 Name: _____

6B · Chester history route: data table

Units are km unless marked. Use + and − signs for components. Row 1 is done for you.

Leg	Given displacement	Length on paper (cm)	x (km)	y (km)	Running R_x	Running R_y	Displacement from start (km)	Direction from start	Total distance so far (km)
1	1.80 km at 34° N of E	7.20	+1.49	+1.01	1.49	1.01	1.80	34° N of E	1.80
2	0.85 km at 69° N of E								
3	1.41 km at 82° N of E								
4	1.97 km at 15° S of W								

Measure your final dashed arrow (Subaru Park to the seminary site) with your ruler and protractor:

	Measured on map	Calculated (from table)
Final displacement (km)	_____ cm × 0.25 = _____ km	
Final direction		

6B · Show your work for each leg

Leg 1 (model). Paper: 1.80 ÷ 0.25 = 7.20 cm
x = 1.80 cos 34° = 1.80 × 0.829 = +1.49 km y = 1.80 sin 34° = 1.80 × 0.559 = +1.01 km
Running: R_x = 1.49, R_y = 1.01 Displacement = √(1.49² + 1.01²) = √3.24 = 1.80 km
Direction = $\tan^{-1}(1.01 \div 1.49)$ = 34°, both + → **1.80 km at 34° N of E**

Leg 2

Paper length · x and y · running totals · displacement · direction

Leg 3

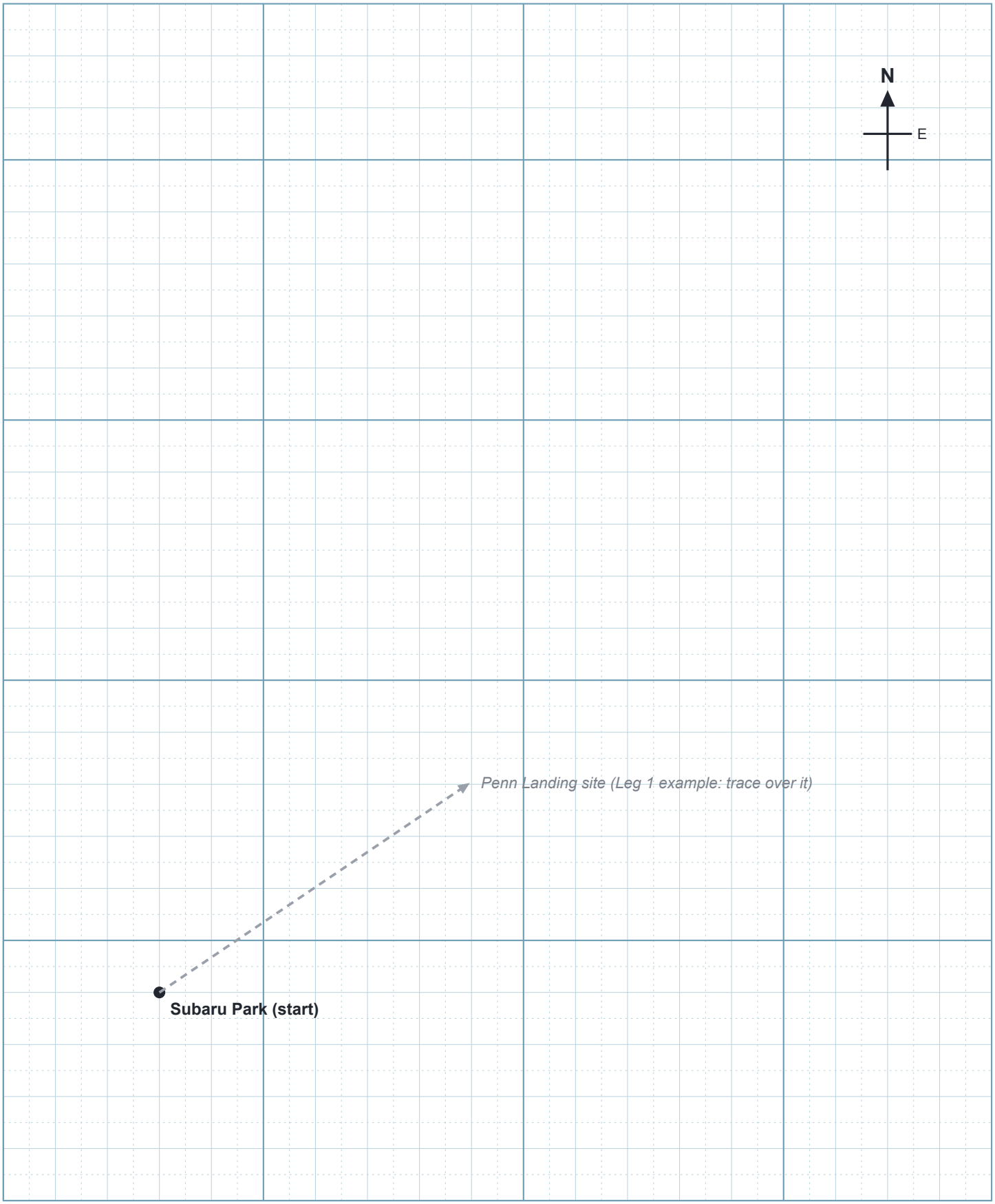
Paper length · x and y · running totals · displacement · direction

Leg 4 Remember: west and south are negative.

Paper length · x and y · running totals · displacement · direction

6B · Chester route map sheet

Name: _____ Scale: 1 cm = 0.25 km



Each small square = 0.5 cm. Each bold square = 5 cm. Print at 100% ("Actual size"). Solid arrows = legs. Dashed arrows (2nd color) = displacement from start.

6B · Questions about your map

Q1 After all four legs, what is your **total distance**? What is your **displacement** from Subaru Park (size and direction)?

Total distance = _____ Displacement = _____

Q2 Why is your displacement so much smaller than your total distance? Point to your map in your answer.

Q3 Find the **percent difference** between your measured and calculated final displacement: |measured − calculated| ÷ calculated × 100. Give one reason they are not exactly equal.

Work

Percent difference = _____ %

Q4 The bus drives straight back to Subaru Park. Draw that leg on your map. What is the displacement for the **whole** round trip? About how far did the bus drive in total?

Round-trip displacement = _____ Total distance ≈ _____

Q5 ★ **Challenge (optional).** Open your drawing compass to your final displacement arrow and draw a circle around Subaru Park. Name a spot on your map that is the same distance from Subaru Park but in a very different direction. What does this show about why direction matters?

6C · Your own route to school (about 40 min)

Now use a real map. Everyone's route is different, so this part is all yours.

1. Open a map of Chester and the area around it (Google Maps, Apple Maps or a printed map). Pick a **start** you know: home, a relative's house, a park, a store or a bus stop. Your **finish** is school.
2. Plan a route with **at least 3 legs**. At least **1 leg must be diagonal** (not straight N–S or E–W). Many Chester streets run at an angle to north, so this happens naturally.
3. Print the map, or trace the important streets onto the 6C graph paper. Draw a north arrow and copy the map's **scale bar**.
4. Draw each leg as a straight arrow. Use the scale bar and your ruler to find each leg's real length. Use your protractor to find each leg's direction.
5. Fill in the 6C data table: components, running totals and displacement after every leg.
6. Draw the final displacement arrow from start to school. Measure it and compare it with your calculation.

> **How to read a scale bar:** measure the bar with your ruler. If a bar labeled 500 m measures 2.5 cm, then 1 cm = 500 ÷ 2.5 = 200 m = 0.2 km. Write that as your scale.

My scale: 1 cm = _____ km Start: _____ Finish: school

Show your work for each leg

Q6 Look at the trip distance your map app shows. Is it closer to your **total distance** or your **displacement**? Why?

★ **Extension (optional):** Measure the straight-line displacement from Subaru Park to Philadelphia City Hall on a map, and estimate its direction. How many km east and how many km north is City Hall from Subaru Park?

 Name: _____

6C · My route: data table

Units are km. Use + and − signs for components. Leave extra rows blank if you don't need them.

Leg	From → To	Length (km)	Direction	x (km)	y (km)	Running R_x	Running R_y	Displacement from start (km)	Direction from start	Total distance (km)
1										
2										
3										
4										
5										

	Measured on map	Calculated (from table)
Final displacement (km)		
Final direction		

6C · My route map

Name: _____ My scale: 1 cm = _____ km

Trace your map here, or tape a printed map on top. Copy the scale bar and mark start and school.



Each small square = 0.5 cm. Each bold square = 5 cm. Print at 100% ("Actual size").

Reflection and submit

About 15 min

R1 Which part was hardest this week? What helped, or what are you still stuck on?

R2 When did your measured and calculated answers disagree the most? What caused it?

R3 Explain to a younger student, in 2 or 3 sentences, why "how far you went" and "how far you are from the start" are not the same. Use your Chester map as the example.

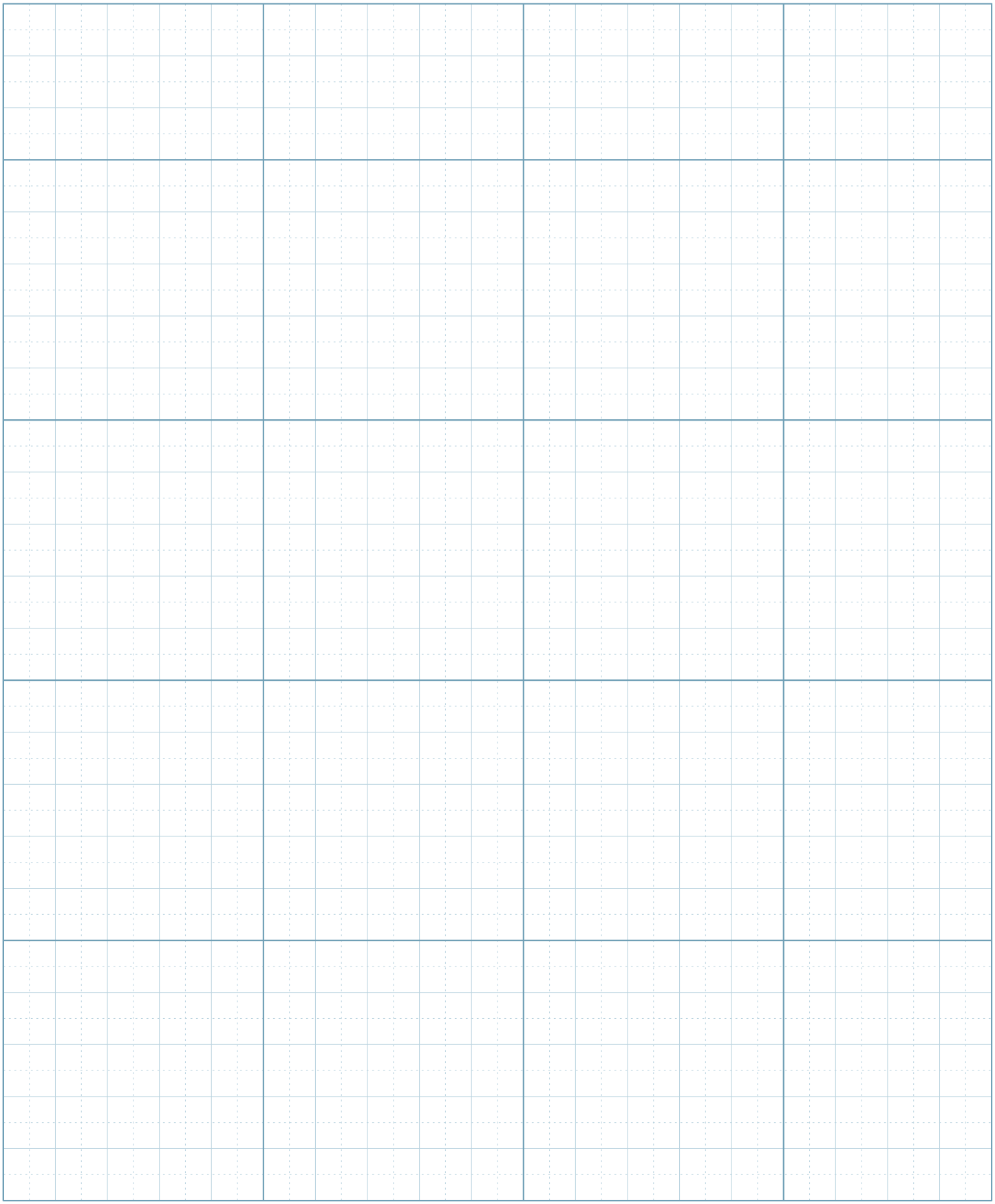
R4 Look at your plan on page 1. Did you stick to it? What would you change next time?

Before you submit

Done	Check
	Every page has my name
	Every calculation shows the formula, the numbers plugged in, and the answer with units
	Every drawing has a scale, a north arrow and labels
	Measured values are written next to calculated values
	Clear, well-lit photo or scan of every page, uploaded **in order** to Canvas

Extra graph paper

Name: _____ Scale: 1 cm = _____



Each small square = 0.5 cm. Each bold square = 5 cm. Print at 100% ("Actual size").